

Criticality Thresholds in One-Dimensional Multiplying Media with n-Bonacci Aperiodic Modulation

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Overview

This deposit contains all code, figures, and manuscript files for the paper:

Nogueira Grossi, P. (2026). *Criticality Thresholds in One-Dimensional Multiplying Media with n-Bonacci Aperiodic Modulation: Spectral Gap Control of k-effective in Substitution-Sequence Diffusion Operators*.

The paper studies the one-group neutron diffusion equation on a uniform 1D slab whose material coefficients vary site-by-site according to the n-bonacci substitution sequence for $n = 2, 3, 4, 5$ (Fibonacci, Tribonacci, Tetranacci, Pentanacci). The central result is that the critical fission strength $\lambda_c(n)$ is governed by the spectral gap of the substitution transfer matrix, not by the n-bonacci constant alone, and that $\lambda_c(n) \rightarrow 7/6$ exactly for $n \geq 4$.

This work is a companion to:

Nogueira Grossi, P. (2026). *Differential Nonlinear Robustness of Critical States in Fibonacci and Tribonacci Substitution Chains*. Zenodo. <https://doi.org/10.5281/zenodo.20026942> (concept DOI, always latest)

Repository structure

```
deposit/
├── README.md           - this file
|
├── nbonacci_criticality.py - pilot sweep: k_eff vs lambda,
|                           Fibonacci and Tribonacci, generates
|                           early diagnostic figures
|
```

— nbonacci_critical_lambda.py	– main sweep: finds $\lambda_c(n)$ for $n=2..5$
	by bisection (Brent's method),
	computes spectral gaps and convergence
	tables, generates fig_lambda_c_v2.png
— generate_all_figures.py	– generates all 5 paper figures
	(Fig 1–5) from scratch; single entry
	point for full reproduction
— nbonacci_diffusion_draft.tex	– LaTeX source of the manuscript
— nbonacci_diffusion_draft.pdf	– compiled PDF of the manuscript
— fig1_chain_structure.png	– Fig 1: substitution word visualization
— fig2_keff_vs_lambda.png	– Fig 2: k_{eff} vs λ , gen 4/6/8
— fig3_lambda_c_gap.png	– Fig 3: $\lambda_c(n)$ vs spectral gap Δ_n
— fig4_convergence.png	– Fig 4: convergence with generation
— fig5_flux_profiles.png	– Fig 5: fundamental flux modes

Reproducing all results

Requirements

- Python ≥ 3.10
- numpy, scipy, matplotlib

Install dependencies:

```
bash
pip install numpy scipy matplotlib
```

Run the full figure pipeline

```
bash
python generate_all_figures.py
```

This reproduces Figures 1–5 exactly as they appear in the paper.

Runtime: under 5 minutes on a single core.

Run the critical lambda sweep (main numerical result)

```
bash

python nbonacci_critical_lambda.py
```

Prints the $\lambda_c(n)$ table, spectral gap data, and convergence tables for $n = 2, 3, 4, 5$ and saves `fig_lambda_c_v2.png`.

Compile the manuscript

```
bash

pdflatex nbonacci_diffusion_draft.tex
pdflatex nbonacci_diffusion_draft.tex    # second pass for references
```

Requires a standard LaTeX distribution (TeX Live, MiKTeX) with the packages: amsmath, amssymb, amsthm, geometry, hyperref, graphicx, booktabs, xcolor, authblk.

Key numerical results (summary)

n	Chain	ρ_n	Δ_n	$\lambda_c(n)$	vs 7/6
2	Fibonacci	1.6180	1.0000	1.06419	outlier
3	Tribonacci	1.8393	1.1019	1.15626	$\approx 37/32$
4	Tetranacci	1.9276	1.1093	1.16667	$\rightarrow 7/6$ exactly
5	Pentanacci	1.9660	1.0949	1.16667	$= 7/6$ (machine)

Linear fit: $\lambda_c \approx 0.958 \cdot \Delta_n + 0.107$ ($r = 0.989$)

Parameter map (dimensionless model)

Symbol	Role	D	Σ_r	$v\Sigma_f$
A (0)	Fissile	1.0	0.5	λ
B,C,...	Absorber	1.0	2.0	0

Vacuum boundary conditions ($\varphi = 0$) at both ends. Uniform mesh, $h = 1$. Harmonic-mean interface diffusion coefficients. Eigenvalue problem solved via `scipy.linalg.eig`; λ_c found by Brent's method (`scipy.optimize.brentq`), tolerance 10^{-9} .

Relation to companion deposit

The DNLS companion paper (concept DOI: doi:10.5281/zenodo.20026942, V4: doi:10.5281/zenodo.20075822) studies nonlinear dynamics on the same Fibonacci and Tribonacci chains using the discrete nonlinear Schrödinger equation. The present deposit extends that spectral framework to classical neutron transport, replacing the DNLS nonlinearity parameter with the fission production strength λ and replacing IPR with k_{eff} as the central observable.

Code for the DNLS companion is available at: <https://github.com/grossi-ops/Atratores/tree/main/DNLS>

Citation

If you use this code or data, please cite:

Nogueira Grossi, P. (2026). Criticality Thresholds in One-Dimensional Multiplying Media with n-Bonacci Aperiodic Modulation [Data and code]. Zenodo. <https://doi.org/10.5281/zenodo.20077205>